Week 24 — Extended Reading: A synthetic matching capstone

STUDY-RL

Source audit: 20 October 2026

Read for a claim, an assumption and a test

Read the teaching guide first, then use this pack alongside the lecture. The annotations below are course-written paraphrases and investigation prompts, not quotations from the papers. Each source link identifies a primary paper or official guidance. Benchmark results do not transfer automatically to our small exercises. For each reading, write the mathematical object, the needed assumptions and one observation that would challenge your interpretation.

1 Kostrikov et al. (2021): IQL revisited

Primary source. https://arxiv.org/abs/2110.06169

Mechanism and scope. For this capstone, focus on the difference between fitting values on logged actions and extracting a policy. The implementation is a small offline warm start, with an intentionally narrow environment and feedback process.

Read, derive, investigate. Identify which examples are absent from the log. Can the learned actor emit decisions never adequately evaluated? Separate a support diagnostic from a formal safety guarantee. The paper's benchmark does not certify the capstone's fairness surrogate.

2 Dudik, Langford and Li (2011): Doubly robust evaluation

Primary source. https://arxiv.org/abs/1103.4601

Mechanism and scope. Doubly robust estimation combines a reward-model prediction with an importance-weighted residual. Correct propensity or reward specification can remove the population bias under the relevant support and sampling assumptions.

Read, derive, investigate. Condition on the context and expand the weighted residual expectation. Which terms cancel with correct propensities? Explain why consistency of a fitted nuisance model is asymptotic and why evaluation rows should be separated from reward-model fitting. SNIPS self-normalisation is not a general proof of lower variance.

3 Hardt, Price and Srebro (2016): Equality of opportunity

Primary source. https://arxiv.org/abs/1610.02413

Mechanism and scope. The paper defines fairness conditions involving predictions, groups and outcome labels, including equalised odds and equality of opportunity. These differ from demographic parity, which conditions only on group.

Read, derive, investigate. Write all denominators before comparing rates. If every applicant is rejected, which parity conditions hold trivially and what utility is lost? Broad claims that demographic parity and equalised odds can never coexist are false; meaningful tradeoff statements require additional premises.

4 Ray, Achiam and Amodei (2019): Constrained benchmarking

Primary source. https://cdn.openai.com/safexp-short.pdf

Mechanism and scope. Use this benchmark to compare reporting practices: reward, expected cost and training-time violations describe different aspects of a constrained learner. A monitoring curve can reveal a failure hidden by a final reward average.

Read, derive, investigate. The capstone actor and dual update use a batch gap in recommendation probabilities, whereas the window metric reports realised decisions. Derive both from the actual arrays and explain why they differ. This policy-level probability surrogate is not the benchmark's additive physical cost channel.

5 MacDiarmid et al. (2025): Reward hacking and generalisation

Primary source. https://arxiv.org/abs/2511.18397

Mechanism and scope. The study investigates reward hacking and associated broader misalignment in particular production-RL experiments. It motivates evaluating behaviours beyond the rewarded task, while preserving the limits of the studied setting.

Read, derive, investigate. Design a capstone proxy that rises when recommendation volume rises even if match quality does not. Hold out an independent outcome measure. The synthetic matching project does not reproduce the paper's model, training environment or behavioural findings.

Connect the reading to your workbook

Complete a one-page capstone report: define the eligible population, the probability-gap objective, realised-rate diagnostics, held-out utility, OPE support, and uncertainty. Distinguish a point estimate from a lower confidence bound and both from a safety guarantee. Save the selected model before final evaluation; do not tune on the evaluation outcomes.

Evidence boundary. Source identities and the central claims above were checked against primary material. These annotations do not reproduce every proof, training run or empirical result. Use the original paper for theorem assumptions and experimental details; use `project/` and the workbook for the exact course implementation.

Statistics reading bridge

A small subgroup can dominate uncertainty in a safety audit. Read interval procedures with each subgroup denominator explicit. Distinguish a zero observed event count from a sufficiently small population rate, and keep the capstone selection and final held-out audit separate.

Checked textbook locations: Siegrist § 8.1, PDF p.960; Ross § 7.3, PDF p.309. https://stats.libretexts.org/@go/page/10200. These locations support the probability and estimation concepts; the algorithm-specific results remain attributed to the primary readings above. The course source catalog distinguishes the original textbook from its supplied instructor manual and records PDF-viewer pagination.